

# AURORA

A Pomodoro Focus Timer with Smart Analytics

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FOCUS

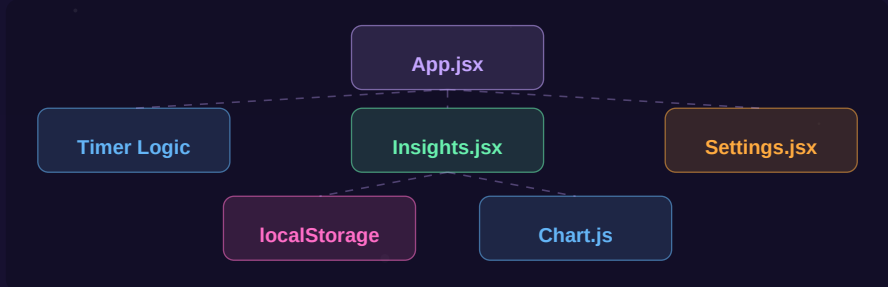
## ABSTRACT

Aurora is a Pomodoro-based productivity application that helps users manage focus and break sessions while providing detailed analytics on their work habits. Built as a modern single-page application, it features an interactive circular timer, a tagging system for categorising work, and a comprehensive insights dashboard with real-time chart visualisations. The app includes browser notification support and is deployed live on Vercel. All data persists locally in the browser, requiring no backend or account setup.

## KEY FEATURES

- Interactive Pomodoro Timer** — Circular SVG progress ring with gradient stroke and glow animation. Configurable focus (default 25m) and break (default 5m) durations with quick presets
- Session Tagging System** — Create and manage project modules to categorise focus sessions. Tags combine module name with custom labels for granular tracking
- Analytics Dashboard** — Real-time bar and pie charts powered by Chart.js showing focus distribution across tags, with time-based filtering (Today / Week / All Time)
- Browser Notifications** — Native browser notification support alerting users when focus or break sessions complete, with permission handling and toggle control
- Persistent Local Storage** — All sessions, settings, and modules persist in localStorage. Zero setup, no account required, complete data privacy
- Live Deployment** — Hosted on Vercel with automatic builds from the repository, accessible at aurora-react-delta.vercel.app

## ARCHITECTURE



## TECH STACK

React 19 Vite 8 Chart.js ESLint CSS Variables Vercel

- React 19.2** with hooks (useState, useEffect, useRef, useLayoutEffect) for all state management
- Vite 8** with Oxc plugin for fast HMR development and optimised production builds
- Chart.js 4.5** for bar and pie chart visualisations with custom aurora theming
- Vercel** for production hosting with automatic deployment from repository

## CONCLUSIONS & FUTURE WORK

Aurora successfully delivers a polished, fully client-side Pomodoro experience with rich analytics and browser notifications. The glassmorphism UI with aurora-inspired animations creates an immersive, distraction-free focus environment. The Settings page is now fully built out with notification controls, module management, and session configuration. The app is live on Vercel for public access.

- Sound alerts alongside browser notifications on timer completion
- Data export/import for backup and cross-device transfer
- Daily and weekly productivity goals with streak tracking
- Break session analytics and extended session history
- Dark/light theme toggle and additional customisation options
- Cloud sync for cross-device session continuity
- PWA support for installable mobile experience

## DESIGN & UI

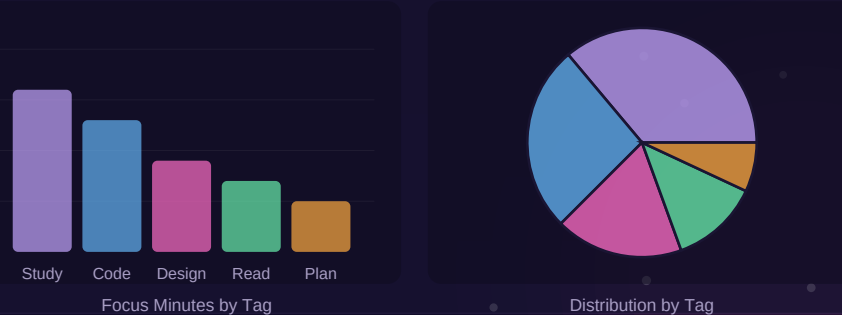
Aurora employs a **dark glassmorphism** design with deep purple backgrounds and lavender accents, creating an ambient, distraction-free focus environment. The interface uses backdrop blur effects, gradient overlays, and animated parallax star fields for visual depth. Typography uses the Inter font stack with responsive sizing across breakpoints.

#120f24 Deep BG #c5a6ff Accent #e9e2ff Text #fff8ef Heading

## KEY METRICS

3 COMPONENTS 535 LINES (APP) 304 LINES (INSIGHT) 158 LINES (SETTINGS)

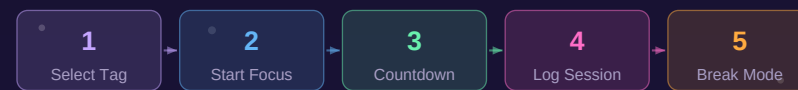
## DATA VISUALISATION



## IMPLEMENTATION DETAILS

- Timer Engine** — useEffect with 1-second intervals and useRef to prevent double session logging. Auto-switches between focus and break modes on completion
- State Persistence** — Initialiser functions read from localStorage on mount; dedicated useEffect hooks sync every state change back. Data survives page refreshes
- Notification System** — Requests browser notification permission on toggle. Fires native notifications when focus or break sessions complete, with localStorage-persisted preference
- Chart Lifecycle** — Chart instances destroyed before recreation on data changes, preventing memory leaks. Custom styling matches the aurora theme palette
- SVG Progress Ring** — Stroke-dashoffset calculations with inline gradients and CSS glow animations for a smooth countdown visual
- Module System** — Case-insensitive duplicate prevention, combined module + tag labelling, and stateful pill-based management UI in the Settings page
- Responsive Layout** — Mobile-first CSS with 768px breakpoint, Inter font stack, and CSS custom properties for consistent theming across all components

## SESSION FLOW



Each completed focus session is automatically recorded with its tag, duration, and timestamp. The insights dashboard aggregates this data in real-time, providing bar and pie chart breakdowns of productivity patterns across customisable time ranges. A browser notification is fired when each session completes.